

Oluwatobi Melvyn Mayungbo

Data Scientist, Credit and Fraud Risk

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SUMMARY

Data scientist with a B.Sc. in Statistics focused on credit decisions. Built a credit decisioning system with calibrated default probabilities, cost-optimal approve and decline cutoffs and reason codes, and an uplift study with experiment design, power analysis and CUPED. Writes SQL and Python.

SKILLS

Payments and Risk: Credit Risk Decisioning, Transaction Fraud Detection, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

Statistics and Causal Inference: Experiment Design, Power Analysis, CUPED, Logistic Regression, WOE Scorecards, Statistical Inference, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting

Data Analysis: SQL, Pandas, NumPy, Anomaly Detection

Engineering and MLOps: Python, SQL, DuckDB, PostgreSQL, Redis, FastAPI, Docker, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

AI and LLM Systems: Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

PROJECTS

Credit Risk Decisioning and Fairness Audit System | [GitHub](#) | [Live App](#)

Tech: Python, LightGBM, scikit-learn, SHAP, WOE Scorecards, FastAPI, Docker

- Developed a full credit decisioning system producing calibrated default probabilities, cost-optimal approve and decline cutoffs, adverse action reason codes and an append-only audit trail that lets any individual decision be reconstructed months later.
- Benchmarked gradient boosting against a traditional WOE-binned logistic scorecard, prioritising calibration quality (Brier score, reliability diagrams, expected calibration error) over ranking metrics, because expected loss needs true probabilities rather than rankings.
- Audited fairness across sex, age, education and marital status using demographic parity and equal opportunity differences plus within-group calibration, and presented the accuracy and fairness tradeoff curve as a policy decision rather than silently optimising it away.

Uplift Modelling for Campaign Targeting | [GitHub](#)

Tech: Python, scikit-learn, LightGBM, EconML, pytest, GitHub Actions

- Ran a causal inference study on the Hillstrom randomised e-mail trial (64,000 customers) to decide which customers to contact under a fixed budget, delivered as a two-page decision memo for a marketing director.
- Evaluated with Qini curves, AUUC, decile tables and bootstrap intervals; uplift targeting was worth about \$1,390 against about \$3,170 for raising the budget, and a failed placebo test on the primary campaign was reported openly.
- Added a power and MDE calculator and a CUPED variance-reduction demo; 135 tests on synthetic data with CI on Python 3.10 to 3.12.

Self-Operating Demand Forecasting Platform | [GitHub](#)

Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions

- Modelled asymmetric business cost by penalising unmet demand at 3× idle supply and shipping the cost-optimal quantile rather than the median, cutting expected cost by 25% on an identical model and beating a seasonal-naive baseline by 32%.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
 - Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.
- Internal Auditor, NISER** May 2024 to July 2024
- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan

Professional Diploma in Artificial Intelligence, SQI College of ICT

January 2021 to March 2025

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action